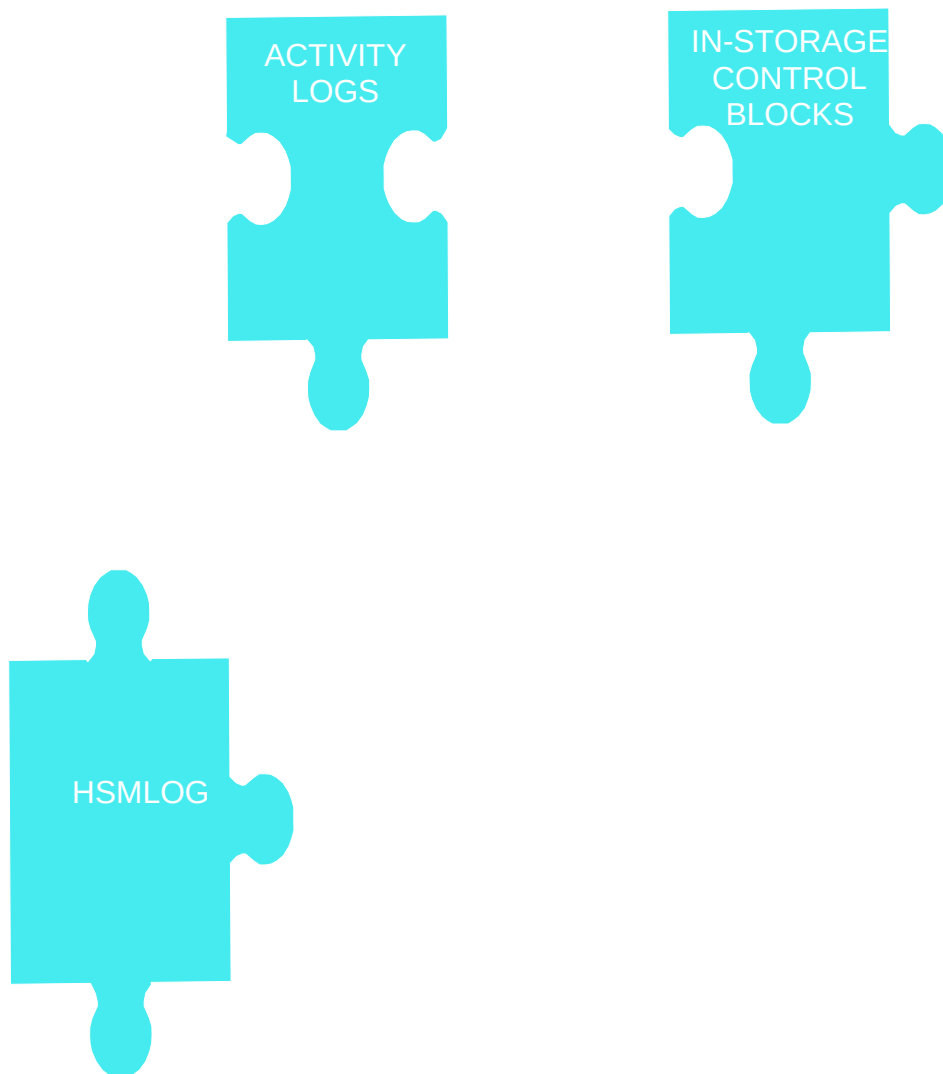
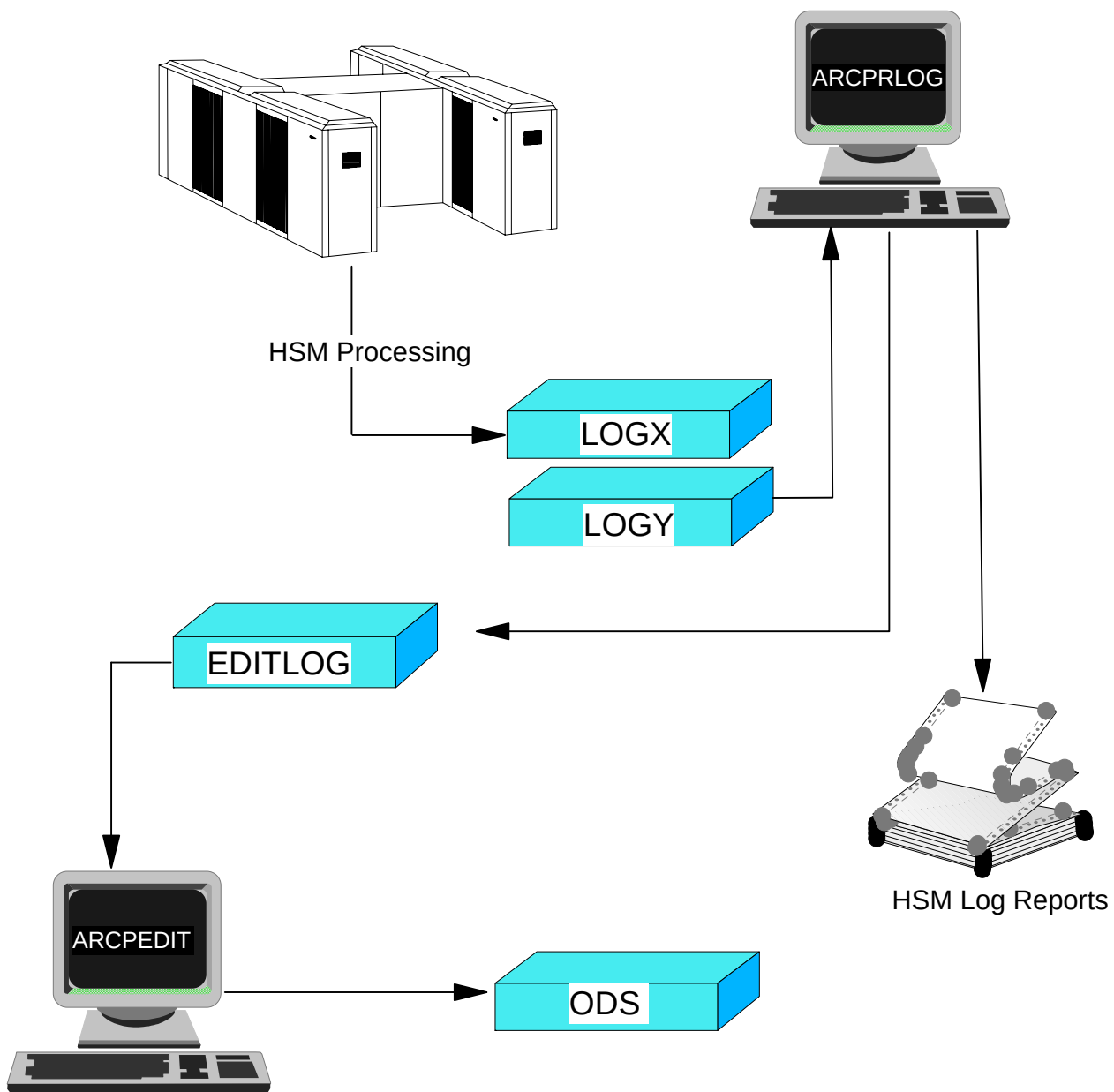






Objectives

- Understand how to implement HSMLOG recording.
- Understand the diagnostic value of the HSMLOG.
- Be able to interpret the HSMLOG data.
- Be able to use the ARCPRLOG and ARCPEDIT to format the raw HSMLOG.



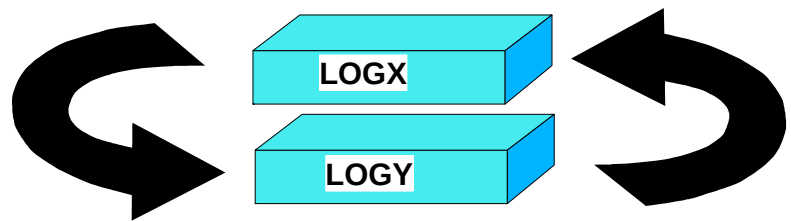
HSM logs are two data sets with which HSM records processing and system event statistics. They are useful for monitoring, tuning, and performing diagnosis of HSM errors.

HSM provides you with two stand-alone programs that can be used to format raw HSMLOG data. In this section we will primarily discuss ARCPRLOG, as it has significantly more diagnostic value than ARCPEDIT. ARCPEDIT provides a 'condensed' version of the output generated by ARCPRLOG. On your own, you may wish to investigate ARCPEDIT as a high-level reporting tool.

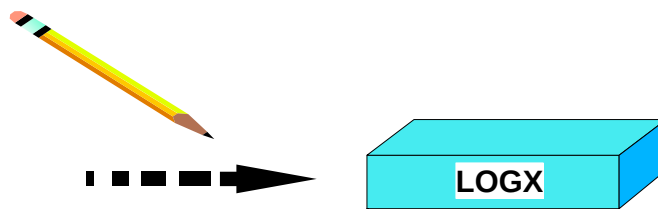
Notes:

Commands Related to the HSMLOG

SWAPLOG



LOG (data)



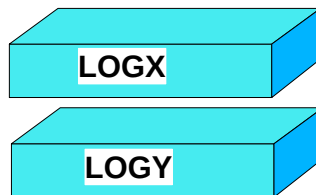
The SWAPLOG command will cause the LOGX to be renamed to the Y and the LOGY to be renamed to the X. Successful completion of the SWAPLOG command will be indicated by a ARC0027I message.

The LOGX and LOGY data sets MUST be placed on the same volume.

The LOG command enters information into the HSM log. Entries are created by the storage administrator, they generally take the form of messages or explanations.

Notes:

HSM Startup Procedure



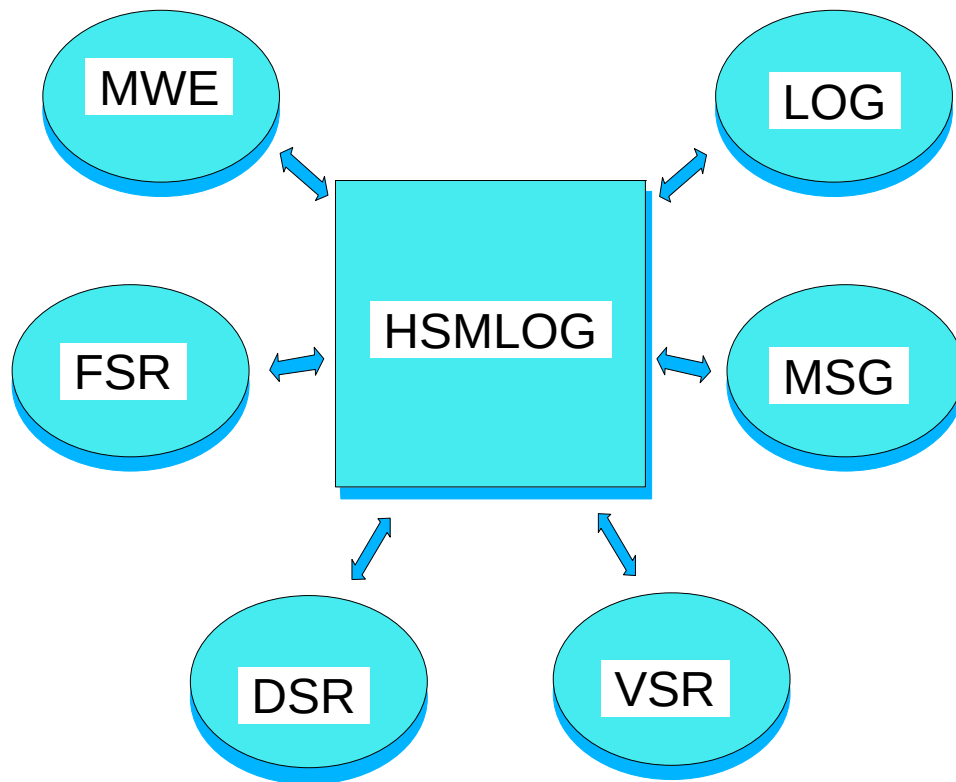
```
HSM  PROC CMD=00,          USE PARMLIB MEMBER ARCCMD00
.....
//      LOGSW=YES,          SWITCH LOGS AT STARTUP
//      UID=DFHSM           DFHSM AUTHORIZED-USER ID
.....
//HSM  EXEC PGM=ARCCTL,DYNAMNBR=DDD,REGION=&SIZE,TIME=1440,
//
PARM=('EMERG=&EMERG','LOGSW=&LOGSW','CMD=&CMD','UID=&UID',
//  'STARTUP=&STARTUP','HOST=&HOST','PDA=&PDA','CDSR=&CDSR',
//  'CDSQ=&CDSQ')
.....
//ARCLOGX DD DSN=&UID..HSMLOGX1,DISP=OLD
//ARCLOGY DD DSN=&UID..HSMLOGY1,DISP=OLD
.....
//ARCPDOX DD DSN=&UID..HSMPDOX,DISP=OLD
//ARCPDOY DD DSN=&UID..HSMPDOY,DISP=OLD
```

Use the LOGSW parameter to swap the log data sets at HSM startup. Swapping the logs at HSM startup will cause HSM to start writing at the beginning of the LOGX data set. HSM always writes to ARCLOGX. When it runs out of space, it 'swaps' the X & Y data sets by renaming them.

HSMLOG data sets are allocated at startup. Each HSM image gets its own set of ARCLOGX/Y data sets.

Notes:

HSMLOG CONTENTS

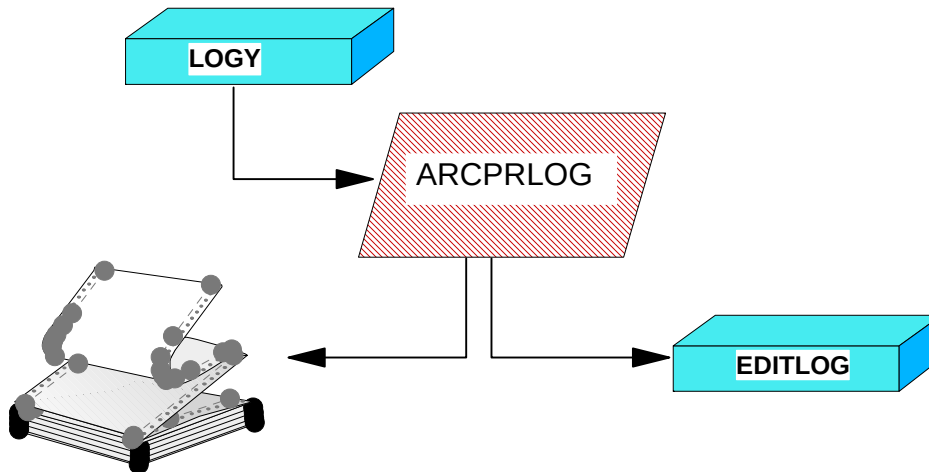


Each log entry is stored as a logical record in a physical sequential log data set. HSM begins each log entry record with a log record header (MLOG) containing the record length, record type, creation time, and creation date. Log entries include the following records:

- MWE tracking statistics gathered at significant points during a data set or volume request. MWE tracking is unique to the HSMLOG.
- FSR records - Functional Statistics Record for a particular function performed against an individual data set.
- DSR records - HSM operating statistics for a day.
- VSR records - Statistics for a particular volume under HSM control.
- HSM error description records - Ex. ARC0208I messages.
- Data entered with the LOG command.

Notes:

ARCPRLOG



```
//HSMLOG JOB
//PRINTLOG EXEC PGM=ARCPRLOG
//ARCPRINT DD SYSOUT=A
//ARCLOG DD DSN=DFHSM.HSMLOGY1,DISP=OLD
//ARCEDIT DD DSN=HSM.EDITLOG,DISP=OLD
//*
//EMPTYLOG EXEC PGM=IEBGENER
//SYSPRINT DD SYSOUT=*
//SYSIN DD DUMMY
//SYSUT2 DD DSN=DFHSM.HSMLOGY1,DISP=OLD
//SYSUT1 DD DUMMY,DCB=(HSM.HSMLOGY1)
/*
```

The ARCPRLOG uses raw HSM log (LOGY) as input and formats the data for printing or further processing by ARCPEDIT. The output of ARCPRLOG is a readable version of the LOG records. Typically, HSM Support will request the raw HSMLOGY data set, instead of the ARCPRLOG formatted log.

- The 'EMPTYLOG' step in the job above nulls the HSMLOGY data set after the 'swap' has been performed. This step is not required and will prevent you from being able to send the raw HSMLOGY data set as documentation for a problem.
- DD statements
 - ARCLOG - Input data set (HSMLOGY).
 - ARCPRINT - Formatted output.
 - ARCEDIT - Edited log data set. Input to ARCPEDIT.

Notes:

Example HSMLOG Output

This is an example of raw HSMLOG.

LEN=0300 TOD=165534 DATE=93/11/23 ID=04 **NAME=ARCCTL RCVE MWE**
MWE FUNC=06 TOD=00:00:00 UID=HSMATH0 **RQN=00000018** ASCB=FA2080
RC=0000 REAS=0000 ABCC=00000000 FLGS=700000 FLG2=0000 FLG3=0000
FLAGS=00 FFLGS=00 MCNT=001 SEQN=0000 CCNT=000 DARC=0000 GROUP=SYS1
RACFLG=81 GEN= 000 BUFL=0000 BUFU=0000 **DSN=HSMTEST.TEST.TEST1**
PW= VSN= DEVT=00000000

LEN=0076 TOD=165534 DATE=93/11/23 ID=07 **NAME=MIGRATION STARTED**
DSN=HSMTEST.TEST.TEST1 FRVSN=PRIM01 TOVSN=MIG001 RC=0000 REAS=0000

LEN=0312 TOD=165535 DATE=93/11/23 ID=08 **NAME=MIGRATION ENDED**
FUNC=MIGRATE L0->L1 TOVOL=MIG001 FRVOL=PRIM01 JOBNM=HSMATH0
RC=00000 REAS=00000 TIMES: REQUEST RECEIVED=165534, STARTED=165534,
ALLOCATED=165534, ENDED=165535. **DSN=HSMTEST.TEST.TEST1** DSORG=PS
RECFM=FB

04EC0000 02FF0000 00000000 0000F3F0 F9F0C8E2 D4C1E3C8 F0400058 D23C0093
327FC8E2 D4C1E3C8 F0400100 C8E2D4E3 C5E2E34B E3C5E2E3 4BE3C5E2 E3F14040
40404040 40404040 40404040 40404040 40404040 40404040 D4C9C7F0 F0F13030
200ED7D9 C9D4F0F1 00000000 00000000 00000000 00000000 0000E2E8 E2F14040
40408100 00000012 0093327F 16553454 16553466 16553572 16553489 0093327F
0093327F 000001E0 00000800 00010001 40006000 00000000 00009000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000

LEN=0077 TOD=165535 DATE=93/11/23 ID=03 **NAME=OUTPUT MSG TO OPER/USER**
MSGID=1000 TO=HSMATH0 NI=2 **ARC1000I HSMTEST.TEST.TEST1 MIGRATE**
PROCESSING ENDED

LEN=0300 TOD=165535 DATE=93/11/23 ID=13 **NAME=MWE COMPLETED**
MWE FUNC=06 TOD=00:00:00 UID=HSMATH0 **RQN=00000018** ASCB=FA2080
RC=0000 REAS=0000 ABCC=00000000 FLGS=500044 FLG2=0000 FLG3=0000
FLGS=00 FFLGS=00 MCNT=001 SEQN=0000 CCNT=000 DARC=0000 GROUP=SYS1
RACFLG=81 GEN=000 BUFL=0000 BUFU=0000 **DSN=HSMTEST.TEST.TEST1**
PW= VSN=PRIM01 DEVT=00000000

This is an example of the output generated using ARCPRLOG.

For complete lists of the printed log fields refer to the following manuals:

DFHSM System Programmer's Guide, Chapter 16
DFSMSHsm Storage Administration Guide for DFSMSHsm, Chapter 17

LEN=0300 TOD=165548 DATE=93/11/23 ID=04 NAME=ARCCTL RCVE MWE
MWE FUNC=03 TOD=00:00:00 UID=HSMATH0 RQN=00000019 ASCB=FA2080
RC=0000 REAS=0000 ABCC=00000000 FLGS=700000 FLG2=0000 FLG3=0000 DFLGS=00
FFLGS=00 MCNT=001 SEQN=0000 CCNT=000 DARC=0000 GROUP=SYS1 RACFLG=81
GEN= 000 BUFL=0000 BUFU=0000 DSN=HSMTEST.TEST.TEST1 PW= VSN=
DEVT=00000000

LEN=0300 TOD=165548 DATE=93/11/23 ID=05 NAME=RECALL/DELETE STARTED
MWE FUNC=03 TOD=00:00:00 UID=HSMATH0 RQN=00000019 ASCB=FA2080
RC=0000 REAS=0000 ABCC=00000000 FLGS=500045 FLG2=0000 FLG3=0000 DFLGS=00
FFLGS=00 MCNT=001 SEQN=0000 CCNT=000 DARC=0000 GROUP=SYS1 RACFLG=81
GEN=000 BUFL=0000 BUFU=0000 DSN=HSMTEST.TEST.TEST1 PW= VSN=
DEVT=00000000

LEN=0312 TOD=165549 DATE=93/11/23 ID=06 NAME=RECALL/DELETE ENDED
FUNC=RECALL L1->L0 TOVOL=PRIM01 FRVOL=MIG001 JOBNM=HSMATH0
RC=00000 REAS=00000 TIMES: REQUEST RECEIVED=165548, STARTED=165548,
ALLOCATED=165548, ENDED=165549. DSN=HSMTEST.TEST.TEST1 DSORG=PS RECFM=FB
DLU=93327 DLM=93327 BYTR=0000002048 TRKR=000001 BYTW=0000000480
TRKW=000001

04EC0000 02FF0000 00000000 0000F3F0 F9F0C8E2 D4C1E3C8 F0400058 D23C0093
327FC8E2 D4C1E3C8 F0400400 C8E2D4E3 C5E2E34B E3C5E2E3 4BE3C5E2 E3F14040
40404040 40404040 40404040 40404040 40404040 40404040 D7D9C9D4 F0F13030
200ED4C9 C7F0F0F1 00000000 00000000 00000000 00000000 0000E2E8 E2F14040
40408100 00000013 0093327F 16554802 16554807 16554921 16554838 0093327F
0093327F 00000800 000001E0 00010001 40006000 00000000 00009000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000

LEN=0077 TOD=165549 DATE=93/11/23 ID=03 NAME=OUTPUT MSG TO OPER/USER
MSGID=1000 TO=HSMATH0 NI=2 ARC1000I HSMTEST.TEST.TEST1 RECALL
PROCESSING ENDED

LEN=0300 TOD=165549 DATE=93/11/23 ID=13 NAME=MWE COMPLETED
MWE FUNC=03 TOD=00:00:00 UID=HSMATH0 RQN=00000019 ASCB=FA2080
RC=0000 REAS=0000 ABCC=00000000 FLGS=500054 FLG2=0000 FLG3=0000 DFLGS=00
FFLGS=00 FLGS=500054 FLG2=0000 FLG3=0000 DFLGS=00 FFLGS=00
MCNT=001 SEQN=0000 CCNT=000 DARC=0000 GROUP=SYS1 RACFLG=81
GEN= 000 BUFL=0000 BUFU=0000 DSN=HSMTEST.TEST.TEST1 PW= VSN=PRIM01
DEVT=3010200E

```

LEN=0163 TOD=081713 DATE=94/05/04 ID=13 NAME=MWE COMPLETED
MWE FUNC=09 TOD=00:00:00 UID=HSMATH4 RQN=00000051 ASCB=FA3D00 RC=0000
REAS=0000 ABCC=00000000 FLGS=500044 FLG2=0000 FLG3=0000 DFLGS=00 FFLGS=00
CMDL=011 CMDO=007 CMDT='SWAPLOG'
LEN=0300 TOD=092302 DATE=94/05/04 ID=04 NAME=ARCCTL RCVE MWE
MWE FUNC=06 TOD=00:00:00 UID=HSMATH4 RQN=00000052 ASCB=FA3D00 RC=0000
REAS=0000 ABCC=00000000 FLGS=740000 FLG2=0000 FLG3=0000 DFLGS=00 FFLGS=00
MCNT=001 SEQN=0000 CCNT=000 DARC=0000 GROUP=SYS1
RACFLG=81 DATE=000000 BUFL=0000 BUFU=0000
DSN=TEST1.CLASS1.EXMAPLE2 pw VSN= DEVT=00000000
LEN=1028 TOD=092302 DATE=94/05/04 ID=14 NAME=DFHSM DAILY STATISTICS
DFHSM DAILY STATISTICS RECORD ----
DATE=94/05/03 NIPL=01 NSHUT=00 NABEND=00 NMWE=0052 CPUSEC=00005
NDSMIGL1=000000 NDSMIGL2=000000 NDSMIGFAIL=0000 NKBYTMIG=000000000
AVGTIME=0000 NDSRECAL1=000000 NDSRECAL2=000000 NDSRECAFAIL=0000
NKBRECA=000000000 AVGTIME=0000 NDUMPSSYS=00000 NDUMPSUSE=00000
NDUMPFFAIL=00000 NDCOPSSYS=00000 NDCOPSUSE=00000 NDCOPFAIL=00000
NVOLRESTSREQ=00000 NVOLRESTFAIL=00000 NDSRESTSREQ=00000 NDSRESTFAIL=00000
10C4E2D9 94123F40 40404040 40404040 40404040 40404040 40404040 40404040
40404040 40404040 40404040 03F81000 00000000 00000000 00000000 00000000
00000001 00000000 00000000 00000034 00000000 000016DA 00000000 00000000
00000000 00000001 00338C60 00000000 00000000 0000C000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
LEN=0090 TOD=092302 DATE=94/05/04 ID=03 NAME=OUTPUT MSG TO OPER/USER
MSGID=1001 TO=HSMATH4 NI=4 ARC1001I TEST1.CLASS1.EXMAPLE2 MIGRATE FAILED,
RC=0034, REAS=0000
LEN=0067 TOD=092302 DATE=94/05/04 ID=03 NAME=OUTPUT MSG TO OPER/USER
MSGID=1234 TO=**OPER** NI=0 ARC1234I LEVEL 2 NOT DEFINED FOR MIGRATION
LEN=0067 TOD=092303 DATE=94/05/04 ID=03 NAME=OUTPUT MSG TO OPER/USER
MSGID=1234 TO=HSMATH4 NI=0 ARC1234I LEVEL 2 NOT DEFINED FOR MIGRATION
LEN=0300 TOD=092303 DATE=94/05/04 ID=13 NAME=MWE COMPLETED
MWE FUNC=06 TOD=00:00:00 UID=HSMATH4 RQN=00000052 ASCB=FA3D00 RC=0034
REAS=0000 ABCC=00000000 <-----MWE ERROR
FLGS=540044 FLG2=0000 FLG3=0000 DFLGS=00 FFLGS=00
MCNT=001 SEQN=0000 CCNT=000 DARC=0000 GROUP=SYS1 RACFLG=81
DATE=000000 BUFL=0000 BUFU=0000 DSN=TEST1.CLASS1.EXMAPLE2 PW= VSN=SCR001
LEN=0163 TOD=092319 DATE=94/05/04 ID=04 NAME=ARCCTL RCVE MWE
MWE FUNC=09 TOD=00:00:00 UID=HSMATH4 RQN=00000053 ASCB=FA3D00 RC=0000
REDEVT=00000000 FLGS=700004 FLG2=0000 FLG3=0000 DFLGS=00 FFLGS=00
CMDL=011 CMDO=000 CMDT='swaplog' AS=0000 ABCC=00000000

```

This is an example of a failed migration request.

Diagnostic value of the HSMLOG

The HSMLOG is generally used to answer the question, "What happened to that recall/recover/backup/migrate/etc. request?" As the previous examples illustrate, the HSMLOG offers an excellent picture of the 'flow' of work within HSM. This flow is accomplished via MWE tracking which is unique to the HSMLOG. This makes it one of the most useful tools for diagnosing performance-related problems. RACF authorization failures are also written to the HSMLOG. This provides an effective means of determining if incomplete requests are the result of RACF violations. Lastly, the HSMLOG provides a quick access to the FSR records.